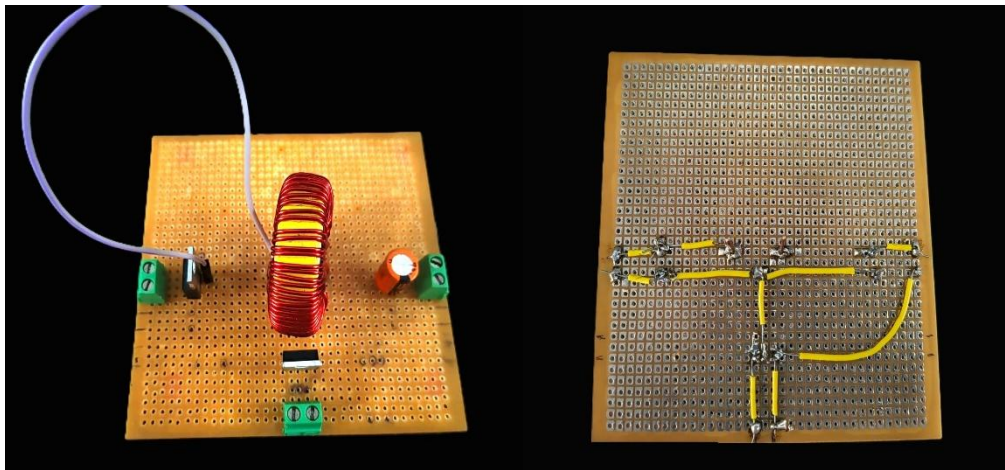


Problem Statement

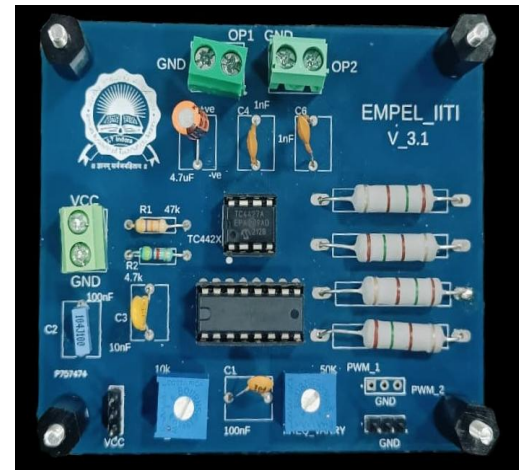
Simulation of a Buck Converter to Observe Waveforms in CCM and Demonstrate DCM by Increasing Load Resistance

Batch B2-G10

Roll no: 240002050, 240002065 , 240002072, 240002083



Soldered Buck Converter Circuit



Gate Driver PCB

We didn't use a heat sink since none of our components were heating up



Inductor Current In CCM (output voltage=16.54V, R=20 Ω)



Inductor Current In BCM (output voltage=17.07V, R=88Ω)



Inductor Current In DCM (output voltage= 20.18, R=150Ω)

In DCM, the inductor current becomes zero before the next cycle, so the diode turns OFF and the switching node loses its normal conduction path. The stray inductances (wires, inductor, PCB tracks) and stray capacitances (MOSFET, diode, probes, layout) then resonate like an LC circuit, producing ringing/distortion. Since the diode voltage, inductor current, and gate signal are all linked to the same switching node through common ground paths and Miller capacitance, the disturbance appears at the same instant in all waveforms.